

Fall 1997

Problem 1 (Fa87, Sp92, Fa97, Sp00, Fa13, Sp19). Define a sequence of real numbers (x_n) by

$$x_0 = 1 \quad x_{n+1} = \frac{1}{2 + x_n} \quad \text{for } n \geq 0$$

Show that (x_n) converges and evaluate its limit.

Problem 2 (Su79, Fa97). 1. Give an example of a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ whose derivative f' is not continuous.

2. Let f be as in Part 1. If $f'(0) < 2 < f'(1)$, prove that $f'(x) = 2$ for some $x \in [0, 1]$.

Problem 3 (Fa97). Let f be an entire function such that, for all z , $|f(z)| = |\sin z|$. Prove that there is a constant C of modulus 1 such that $f(z) = C \sin z$.

Problem 4 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21).

Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1 + x^{2n}}$, where n is a positive integer.

Problem 5 (Fa81, Sp89, Fa97). Let f be a holomorphic map of the unit disc $\mathbb{D} = \{z \mid |z| < 1\}$ into itself, which is not the identity map $f(z) = z$. Show that f can have, at most, one fixed point.

Problem 6 (Fa97). Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be distinct real numbers. Show that the n exponential functions $e^{\alpha_1 t}, e^{\alpha_2 t}, \dots, e^{\alpha_n t}$ are linearly independent over the real numbers.

Problem 7 (Fa97). Define the *index* of a real symmetric matrix A to be the number of strictly positive eigenvalues of A minus the number of strictly negative eigenvalues. Suppose A , and B are real symmetric $n \times n$ matrices such that $x^T A x \leq x^T B x$ for all $n \times 1$ matrices x . Prove the index of A is less than or equal to the index of B .

Problem 8 (Fa97, Fa08). Suppose N_i is a normal subgroup of a group G for $1 \leq i \leq k$, such that $N_i \cap N_j = \{e\}$ for $i \neq j$. Prove that G contains a subgroup isomorphic to $N_1 \times N_2 \times \dots \times N_k$ if $k = 2$, but not necessarily if $k \geq 3$.

Problem 9 (Fa97, Sp09, Sp16, Sp26). Let G be a finite group.

1. If G is non-abelian, show that $G/Z(G)$ is not cyclic, where $Z(G)$ is the center of G .
2. If $|G| = p^n$, with p prime and $n > 0$, show that $Z(G)$ is not trivial and deduce that G has a normal subgroup of order p .
3. If $|G| = p^2$, show that G is abelian.

Problem 10 (Fa97). Prove that for all $x > 0$, $\sin x > x - \frac{x^3}{6}$.

Problem 11 (Fa84, Fa97). Let f be a C^2 function on the real line. Assume f is bounded with bounded second derivative. Let

$$A = \sup_{x \in \mathbb{R}} |f(x)| \quad \text{and} \quad B = \sup_{x \in \mathbb{R}} |f''(x)|$$

Prove that

$$\sup_{x \in \mathbb{R}} |f'(x)| \leq 2\sqrt{AB}$$

Problem 12 (Fa97). A map $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *proper* if it is continuous and $f^{-1}(B)$ is compact for each compact subset B of $\mathbb{R}^n$; f is *closed* if it is continuous and $f(A)$ is closed for each closed subset A of $\mathbb{R}^m$.

1. Prove that every proper map $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is closed.
2. Prove that every one-to-one closed map $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is proper.

Problem 13 (Fa97). Conformally map the region inside the disc given by $\{z \in \mathbb{C} \mid |z - 1| \leq 1\}$ and outside the disc $\{z \in \mathbb{C} \mid |z - 1/2| \leq 1/2\}$ onto the upper half-plane.

Problem 14 (Fa97, Fa10, Sp24). Evaluate the integral $\int_{-\infty}^{\infty} \frac{\cos kx}{1 + x + x^2} dx$, where $k \geq 0$.

Problem 15 (Sp89, Fa97, Fa04). Let S be the subspace of $M_{n \times n}$ (the vector space of all real $n \times n$ matrices) generated by all matrices of the form $AB - BA$ with A and B in $M_{n \times n}$. Prove that $\dim S = n^2 - 1$.

Problem 16 (Fa97). Prove that if A is a 2×2 matrix over the integers such that $A^n = I$ for some strictly positive integer n , then $A^{12} = I$.

Problem 17 (Fa88, Fa97, Fa24). Let the group G be generated by two elements, a and b , both of order 2. Prove that G has a subgroup of index 2.

Problem 18 (Fa97, Fa22). A finite abelian group G has the property that for each positive integer n the set $\{x \in G \mid x^n = 1\}$ has at most n elements. Prove that G is cyclic, and deduce that every finite field has cyclic multiplicative group.